PlayMetrix

Team Pesado

Human Factors in Design - CA1

Agenda

1. Problem Overview
2. Cognition
3. Physical Ergonomics
4. Human Error & Performance
5. Accessibility
6. UI Redesign

Problem Overview

Using the application in a match day setting

Cognition

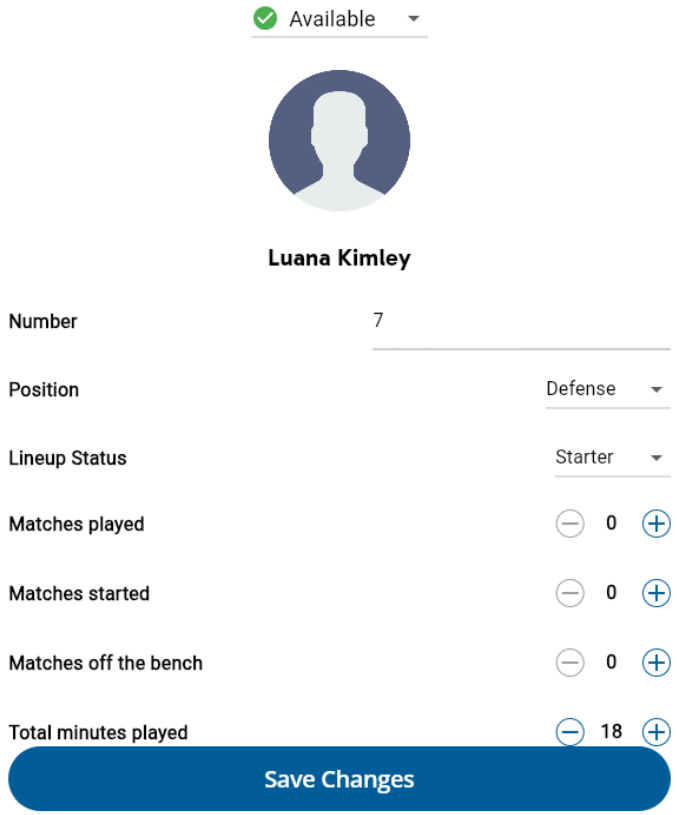
John the Manager calculating duration of each player during a match.

10 steps to input duration

Duration = (Time ended - Time started) + current duration

Cognitive overload

GOMs



Physical Ergonomics

Refined Problem Definition: Attempting to use the application in real time to report an injury as it happens

HFI Issues: Weather, Position around the Pitch, Device getting wet

Findings: App could be used during the match, but the HF issues impacted how successfully/quickly it was done

Moving Forward to CA2: Comparing different devices to see which is the easiest to use

Match is being played in overcast rainy conditions

Human Error

Problem: Physios' assessing and recording an initial player injury during a live match.

- **Slip**: Physio accidentally records a player injury with the wrong date for an injury assessment.
- **Lapse**: The physio forgets to update the injury status of a player.
- **Mistake**: The physio misidentifies a player and documents an injury under the wrong player's name.

Consequences:

- Player health may suffer.
- GDPR breaches.
- Physios not trusting the application.

sportsfile 29 November 2020; Aaron Gillane of Limerick is tended to for a back injury by team physio Mark Melbourne, left, and team doctor Dr James Ryan during the GAA Hurling All-Ireland Senior Championship Semi-Final match between Limerick and Galway at Croke Park in Dublin. Photo by Piaras Ó Mídheach/Sportsfile

Performance

- It takes Sarah 5 button steps in our application to start recording an injury.
- Sarah clicks on Players.
- Sarah clicks on the injured player from the list of 30-40. This may involve scrolling to find the player.
- Sarah has to click on edit player to find the add injury button.

Recommendations for CA2

- Confirmation Prompt for Injury Reporting
- Show Physios the last reports they created to review them
- Reminders for Follow-up and Report Upload

Accessibility

Problem: A manager using the app checks for player availability who suffers from various visual impairment issues such as CVD (color blindness)

Findings: On average 1 and 12 men worldwide suffer from CVD

Types of CVD	Colors Affected
Protanomaly	Red
Protanopia	Red, Green (some shades)
Deuteranomaly	Green, Red
Deuteranopia	Green, Red
Tritanomaly	Blue, Yellow
Tritanopia	Blue, Yellow
Monochromacy (Achromatopsia)	All colors are perceived as shades of gray

Going Forward For CA2

Would like to visualize how someone with each type of CVD perceives different colors, especially when it comes to perceiving our player availability traffic light system

Implement CVD filter settings to allow for our app to be accessible for users with various types of CVD conditions

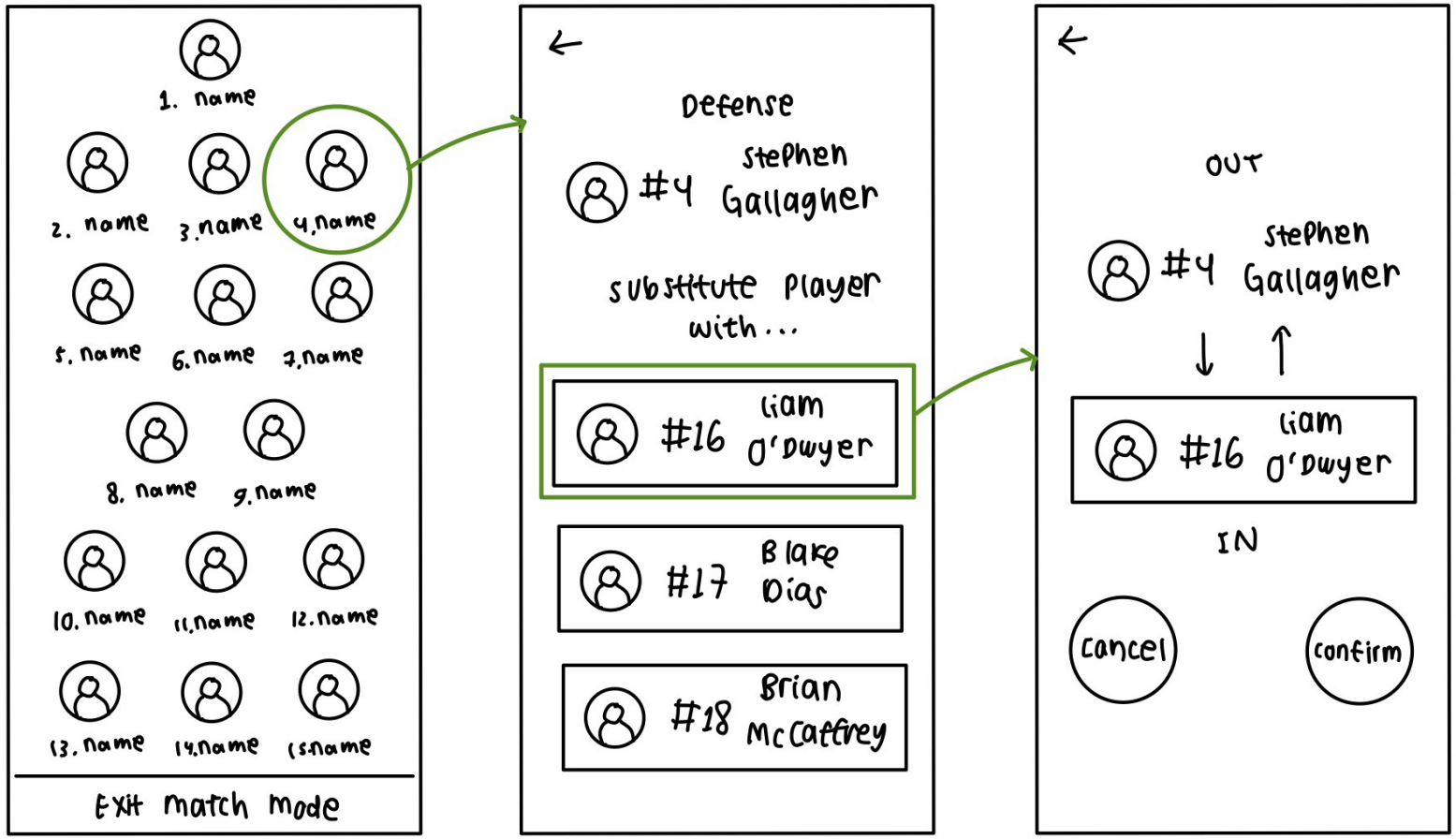
Our solution:

Implementing

“Match Mode”

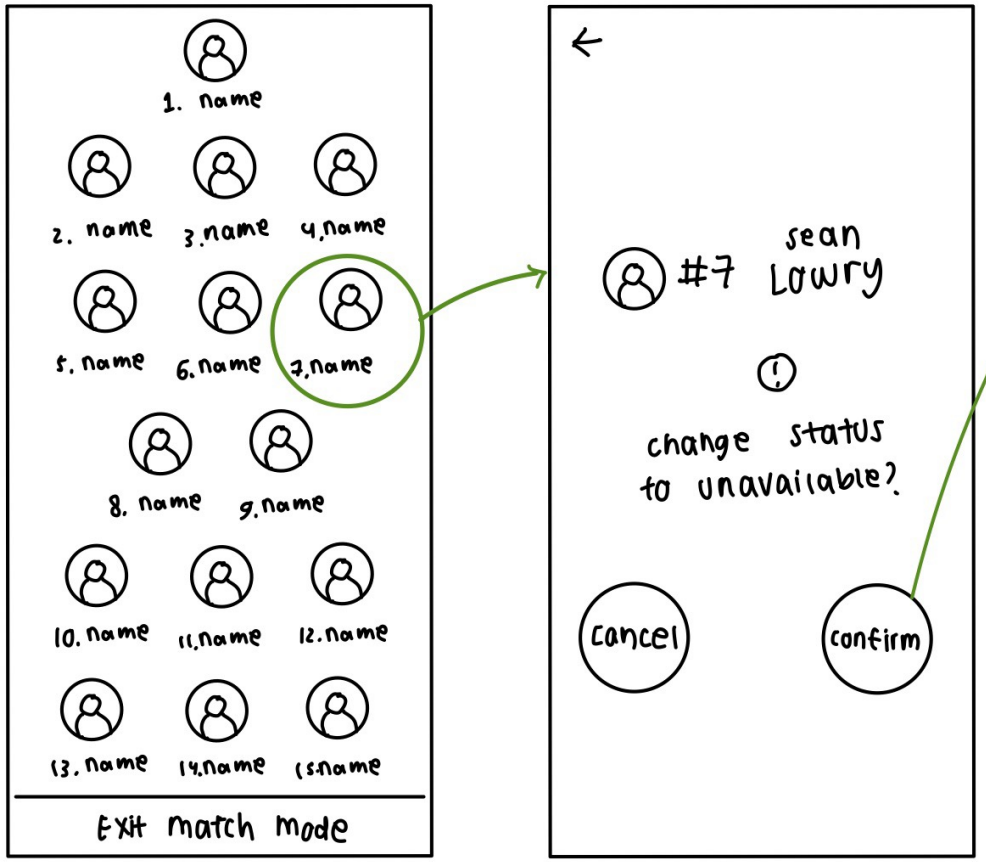
Match Mode - Manager View

- Recording Duration Played



Match Mode - Physio View

- Flagging Injured player



- will send notification to manager / coach

(!) sean Lowry (#7) is unavailable since 7/3/2024 18:00.

- will send notification to physio

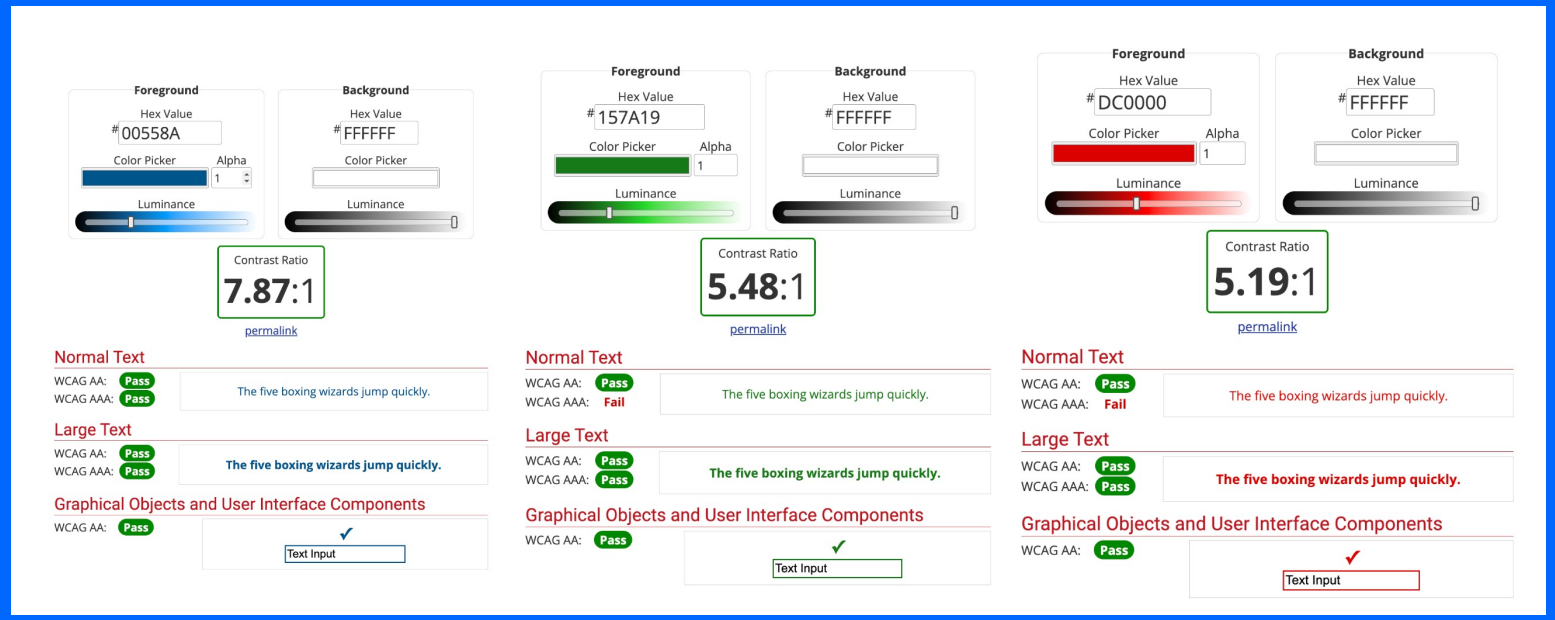
(!) 7/3/2024 18:00 schedule appointment with Sean Lowry

→ Reminder for them to schedule an appointment with the player to further check the injury

Look & Feel

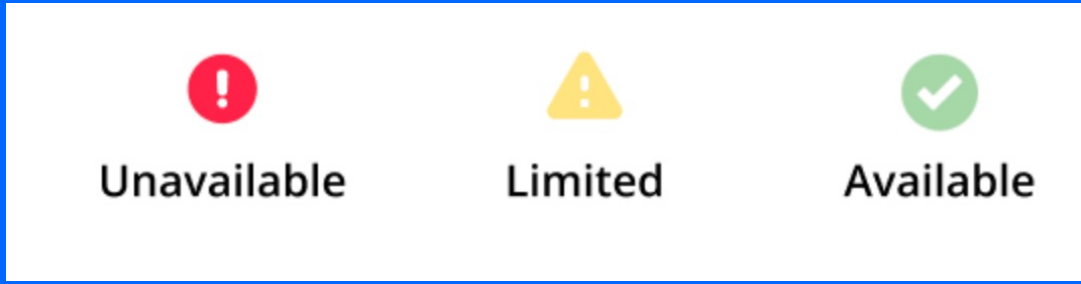
Contrasting colours

- https://webaim.org/resources/contrastchecker/

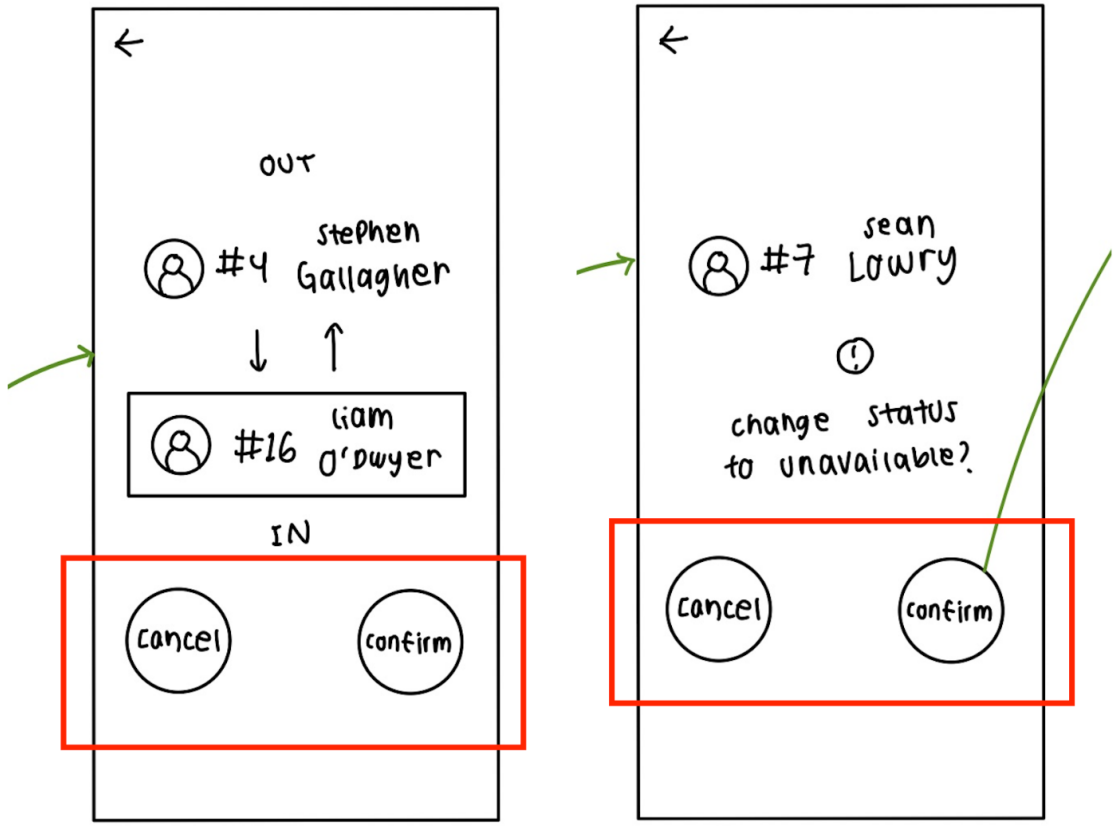


Incorporating icons

- Gives more context especially for people with colour blindness



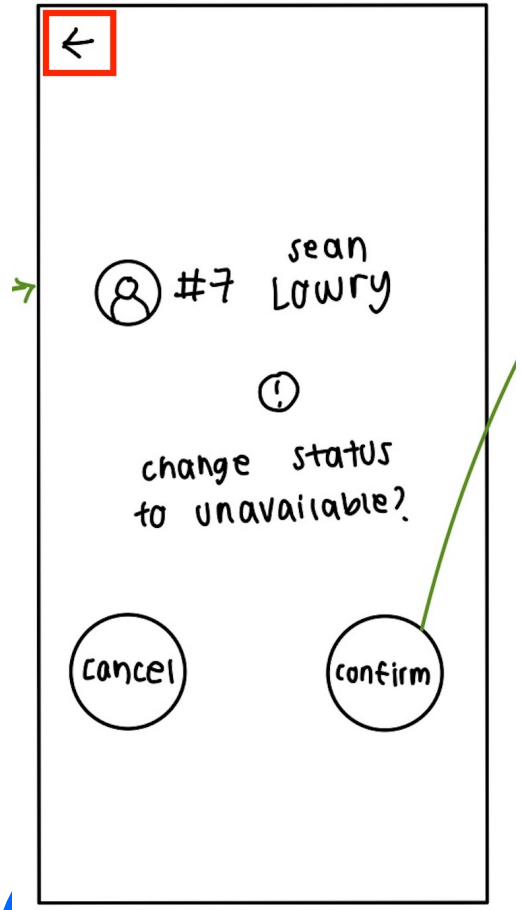
Gestures



Button positions

- Mimicking natural hand movements
- Easy access

Gestures



Back navigation

- Top left of page
- Sliding right to go back

Thank you

Any Questions?

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